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The Role of Gut Microbiome Diversity In the Relationship Between Carbohydrate Quality and Gut-Brain Axis Signaling In Individuals With Bulimia Nervosa

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Research Problem

Unfortunately, current statistics on bulimia nervosa are likely underestimated and do not capture the full picture of the physical, emotional, and social damage it brings [1]-[3]. The reality is that many cases remain undiagnosed due to two major reasons: delayed clinical recognition and discouraged help-seeking [4]. Therefore, the underdiagnosis of bulimia nervosa leads to an underrepresentation of those affected, limiting studies to gain a full understanding of the disorder. Unlike other forms of eating disorders such as anorexia, the extreme restriction of nutrient intake relative to requirements that leads to significantly low body weight that present obvious signs of weight change [5], those with bulimia often have a maintained body weight, appearing physically stable from the outside despite having constant binge-eating and compensatory behaviors [6]. This increases the chance of delayed recognition and explains why bulimia nervosa goes undetected by family members and healthcare providers in many cases.

Furthermore, the societal stigma surrounding this issue discourages patients from seeking diagnosis and professional support, specifically due to denial and shame [7]. Previous research by Wall and colleagues [7] also suggests that eating disorders are especially underdiagnosed within the male population due to the stigma of eating disorders being a predominantly female struggle, leading many to avoid reaching out. According to Wall and colleagues [7], individuals struggling with eating disorders in general commonly fold under the fear of their condition being dismissed as “not a real illness,” further preventing them from seeking help.

While there has been sufficient research on bulimia nervosa and its psychological aspects [7], others have also discovered how the quality of dietary carbohydrate consumed constantly influences the gut microbiome composition, and how the gut microbiome communicates with the brain through the gut-brain axis [8]. However, there is extremely less research that integrates these three areas to determine how gut microbiome diversity influences the relationship between carbohydrate quality and gut-brain axis signaling specifically in individuals with bulimia nervosa. This gap significantly limits the current understanding of the disorder from a biological aspect and provides the basis for the literature review.

Despite growing evidence that dietary carbohydrate quality influences gut microbial composition and that the gut microbiome communicates with the brain through the gut-brain axis, these relationships have largely been investigated independently. Likewise, although bulimia nervosa is known to involve inconsistent and unbalanced nutritional intake and has to have a causal relationship with altered gut microbial diversity, little research has investigated how these factors interact as a whole. As a result, significantly more research on the psychological and behavioral aspects of the condition has been conducted than on the biological consequences. There has been growing evidence that dietary carbohydrate quality influences gut microbiome diversity and gut-brain axis signaling; however, there are limited findings on how these mechanisms interact in individuals with bulimia nervosa.

Existing Literature

Mora-Flores and colleagues [9] reviewed current evidence in reviews of recent human and animal intervention studies on how dietary carbohydrate intake can influence the gut microbiome. Across the sources, the researchers consistently encountered findings of different

carbohydrate structures leading to the growth of certain bacterial taxa. These findings suggest that the quality of dietary carbohydrate ingested likely influences the changes in microbial composition through the production of fermentable substrates and their interactions with intestinal microorganisms.

In 2019, Johnson and colleagues [10] studied 34 adult subjects across a longitudinal study to explore the individual differences in the relationship between diet and the respective gut microbiomes. The results indicated that 78% of the participants showed significant associations between changes in dietary intake and altered microbiome composition. The researchers therefore suggested that diet-microbiome relationships were significantly personalized and differed across the selected population, suggesting that similar dietary intake may produce different responses from the gut microbiomes.

Nireeksha and colleagues [11] discovered that functional diversity in the gut is much more significant than simply just the microbial species in producing the metabolites and signaling molecules that communicate with the brain. Reinforcing these findings, Xiong and colleagues [12] explain that short-chain fatty acids (SCFAs) are microbial metabolites that serve as signaling molecules in the gut-brain axis, produced as dietary fiber is fermented in the gut microbiome. Similarly, Mansuy-Aubert and Ravussin [13] suggested evidence that short-chain fatty acids also activate vagal neurons and encourage enteroendocrine hormones, such as GLP-1 and cholecystokinin, to send signals, further promoting gut-brain communication.

Walsh and colleagues [14] investigated gastric function in women with bulimia nervosa to study whether the disorder alters the stomach's physiological responses to food. The researchers indicated that post-meal gastric relaxation was significantly lower in participants with bulimia nervosa than in the controlled subjects, with an average relaxation of 29.7 mL compared with 105.1 mL. As a result, Walsh et al. [14] indicated that repeated binge eating and vomiting may alter the normal physiological reactivity of the upper gastrointestinal tract, and that altered gastric relaxation, with changes in gastric emptying could interfere with the normal development of fullness, which could contribute to the continuity of binge-purge actions.

A study by Ayton and colleagues determined a strong correlation between the frequent consumption of ultra-processed food sources and individuals diagnosed with bulimia nervosa [15]. Approximately 72% of the diet was classified as ultra-processed foods (NOVA-4) amongst the 73 adult participants with the eating disorder, who excessively consumed foods such as chocolate, ice cream, cakes, and pizza during binge-eating episodes. Although the study did not directly measure the gut microbiome, its observations suggest that the common dietary patterns among those struggling with bulimia nervosa may correlate with more disruptions in the gut microbiome, and therefore decreased gut-brain axis signaling.

Research Question

How and to what extent does gut microbiome diversity influence the relationship between carbohydrate quality and gut-brain axis signaling in individuals with bulimia nervosa?

Methodology

Rather than conducting a hands-on experiment, this study will serve as a transdisciplinary literature review of the fields of psychology, neuroscience, health, and education to further understand how differences in gut microbiome variability can influence the relationship between carbohydrate source quality and gut-brain axis signaling in individuals with bulimia nervosa. Mostly peer-reviewed articles published in the last six years will be selected to ensure that the content was most up-to-date through the latest facts and discoveries. Additionally, major search engines and databases such as Google Scholar, PsychInfo, HOLLIS, and JSTOR will be used to collect this relevant literature.

The literature studied will be categorized in four interconnected topics: (1) how carbohydrate quality affects the gut microbiome, (2) the relationship between gut microbiome diversity and gut-brain axis signaling, (3) the biological consequences of bulimia nervosa on nutrition, the gut microbiome, and the gut-brain axis, and (4) dietary carbohydrate quality in individuals with bulimia nervosa. Respectively, a group of four sets of search terminologies will be used to find fitting sources to pursue the literature review. The first set includes terms such as "carbohydrate quality and gut microbiome", "whole food vs. highly processed carbohydrates", "dietary fiber and gut microbiome", and "refined carbohydrates and gut microbiome fermentation". The second set, which will explore the relationship between gut microbiome diversity and gut-brain axis signaling, will be composed of "gut microbiome diversity and gut-brain axis", "short-chain fatty acids and gut-brain signaling", "gut microbiota and vagus nerve", and "microbial metabolites and brain signaling". Thirdly, the next set of terms that will focus on the biological consequences of bulimia nervosa on the gut-brain axis consists of "bulimia nervosa and gut microbiome diversity", "bulimia nervosa and gut dysbiosis", "bulimia nervosa and gastric emptying", and "bulimia nervosa and gut-brain axis". Finally, the fourth set places attention on dietary carbohydrate quality in individuals with bulimia nervosa and therefore, includes the terms "bulimia nervosa and ultra-processed foods", "bulimia nervosa and dietary fiber", "bulimia nervosa and carbohydrate quality", and "bulimia nervosa and dietary patterns".

During the research process, references will be organized using Zotero, a citation management software. Moreover, the snowballing method will help with discovering additional relevant literature by examining the reference lists and studies cited in the peer-reviewed articles. By using this process, works that may not have appeared through keyword searches alone will be discovered, further widening my research toolbox.

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